Project Report

Project Title

Snake and Ladder Game (GUI) using C++ and Qt

Group Members

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Submission Date

11-May-2025

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1. Executive Summary

• Overview:
This project aims to develop a GUI-based version of the classic Snake and Ladder board game using C++ and Qt. The game applies OOP concepts to build interactive and modular game components.

• Key Findings:
So , the architecture of the game has been established. The GUI board is functional, and the dice changing mechanism and basic movement logic have been successfully implemented. Initial playtesting has confirmed the interface responsiveness and game logic accuracy.

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2. Introduction

• Background:
The Snake and Ladder game project demonstrates core OOP principles such as encapsulation, and inheritancen a fun and interactive way. Developing the game using Qt helps reinforce GUI design.

• Project Objectives:

- ✧ Design and implement a modular game structure using C++ classes.
- ✧ Build an intuitive GUI using Qt.
- ✧ Enable two to four-player functionality and automatic turn handling.
- ✧ Provide with animations and dice rolls.

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3. Project Description

• Scope:
Included:

- ✧ Player movement
- ✧ Board GUI display
- ✧ Dice rolling logic
- ✧ Basic game rules
- ✧ Advanced animations or sounds
- ✧ Home/Start-up Page

• Technical Overview:

- ✧ **Programming Language:** C++
- ✧ **Framework/IDE:** Qt Creator
- ✧ **Graphics:** Basic PNG images for tokens and board

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4. Methodology

• Approach:
We have used an iterative development method with regular testing and integration. Tasks will be modularized based on logic, and integration components.

• Roles and Responsibilities:

Talha : GUI design and Integration and testing of Game Screen.

Abdullah : GUI design and Integration and testing of Home/Start-up Screen

Rameel : Logic implementation of the game

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5. Project Implementation

• Design and Structure:
The project includes the following class hierarchy:

- ✧ **GameController** – manages the overall game state
- ✧ **Player** – tracks position and turn
- ✧ **Dice** – handles random number generation
- ✧ **Board** – maintains the game map with snakes and ladders

The interface has a grid-based layout for the board and control buttons for rolling dice.

• Functionalities Developed (50% complete):
✓Dice rolling and animation
✓Player movement on dice roll
✓GUI layout with board display
✓Turn-based two to four-player logic
✓Snakes and ladders logic
✓Game end detection
✓Sound effects and animations

• Challenges Faced:

- ✧ Difficulty aligning player tokens on the board grid in the correct order
- ✧ Debugging signal-slot connections in Qt
- ✧ Managing real-time movement of tokens (partially resolved with helper methods)
- ✧ Working with GUI and multiple libraries
- ✧ Faced too many errorss

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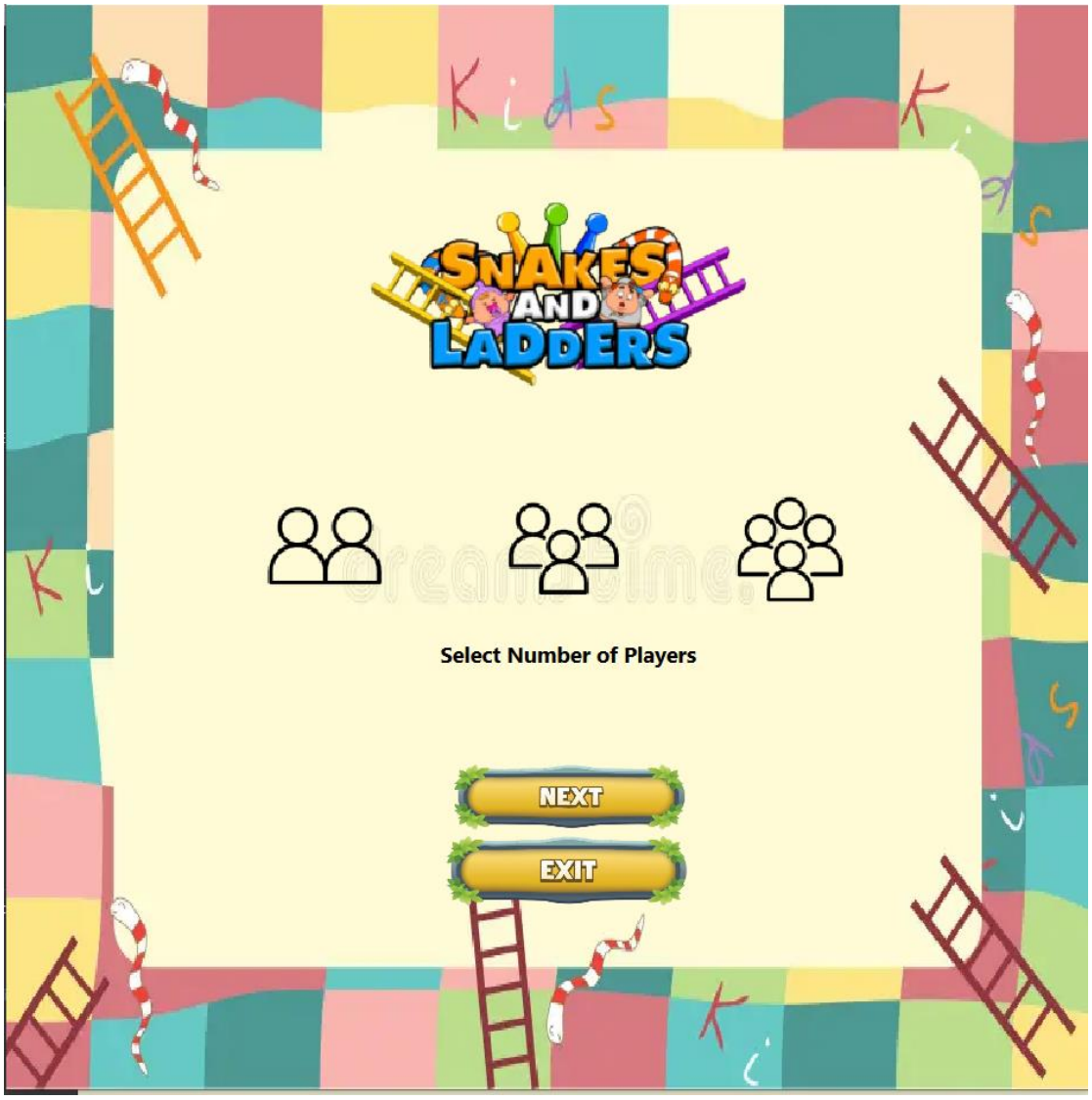
6. Results (So Far)

• Project Outcomes:

- ✧ Functional board display and basic dice mechanics
- ✧ Players can take turns and move tokens
- ✧ GUI is responsive and handles basic interactions well

• Screenshots and Illustrations:

HOMEPAGE



SETUP-PAGE

Kids
Player 1 Name
Player 2 Name
Player 3 Name
Player 4 Name
START
BACK

GAMEPAGE

• **Testing and Validation:**

✧ Manual testing for dice rolls and player movement
✧ Debug logs verify token position updates and turn switching

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7. Conclusion

• **Summary of Findings:**
The project successfully applies OOP concepts and Qt GUI techniques. The dice logic, game turn management, and GUI have been implemented. Key areas like Homepage, datasetupPage and game restart options are planned for completion in the next phase.

• **Final Remarks:**
Progress is steady, and the structure supports adding more features easily. The

project was a great collabrative learning experience for us as it enlightened us with new knowledge and allowing us to learn new skills and implement then accordingly.